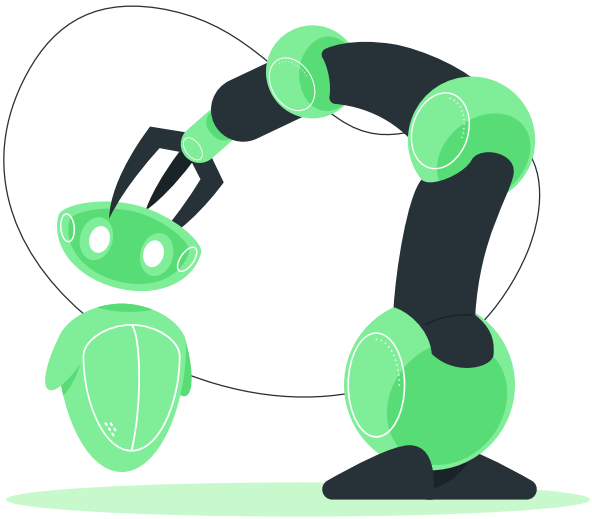
Robotics
CS 3015

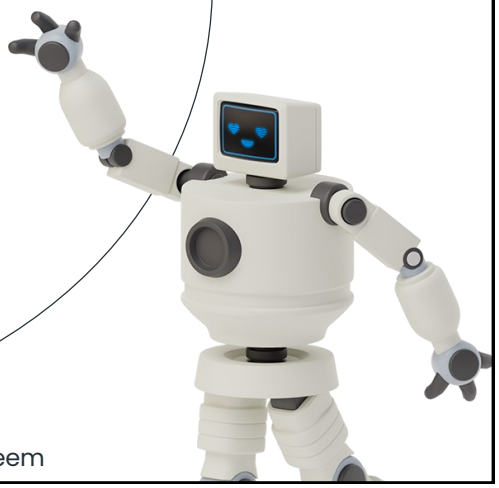
Umarah Qaseem



1

- ROS
- ~~Steps for Linux Installation~~
- Ubuntu, Debian
- Steps for ROS Installation, Setup and Configuration
- How?
- Why?
- When?
- ROS1 and ROS2
- Nodes
- Topics
- Publisher/Subscriber
- Service
- Actions
- ~~File Bag~~
- Packages

15



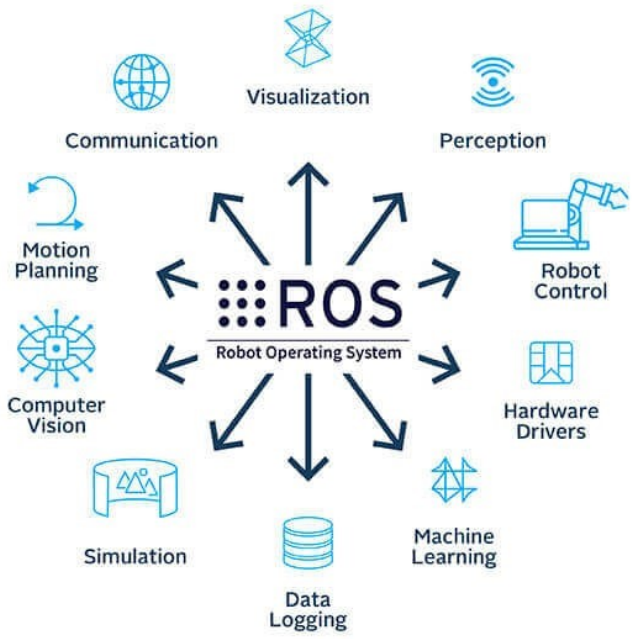
2

Jump Start to ROS Robotic Operating System



Robotics – Umarah Qaseem

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ROS Stands for

- Robotic Operating System

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What is it?

- Not an Operating System!

7

What is it?

- Set of tools and software libraries that will help you build robotic applications

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What is it?

- It is an open-source robotics middleware suite.
- It is a set of software frameworks for robot software development.
- It provides services designed for a heterogeneous computer cluster such as hardware abstraction, low-level device control, implementation of commonly used functionality, message-passing between processes, and package management.

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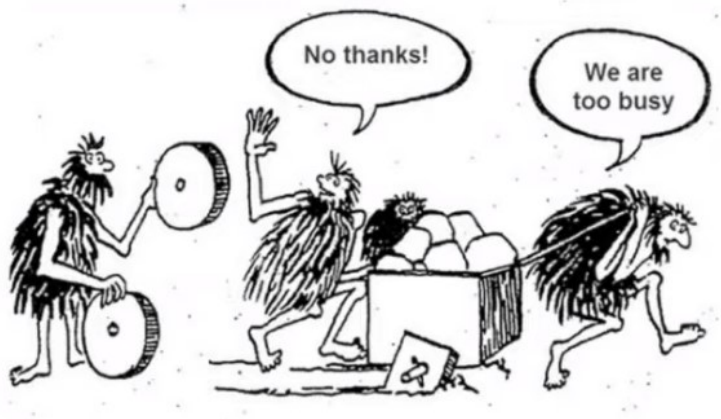
What does it do?

- Provides a standard for robotics applications
- Use on any robot
- Modularity

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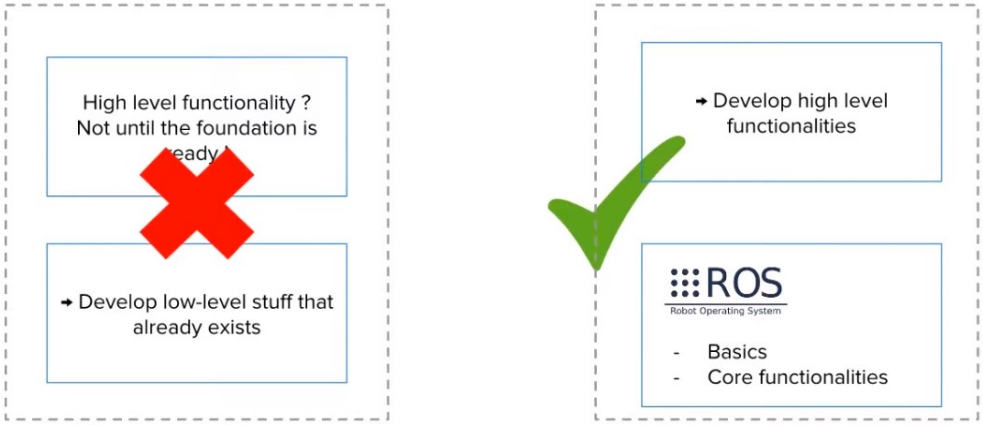
ROS Goals

- Don't reinvent the wheel!



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Why?

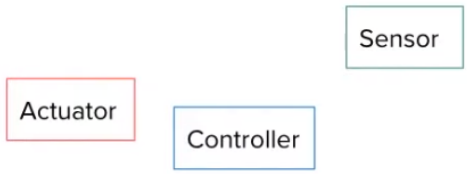


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When to use ROS?

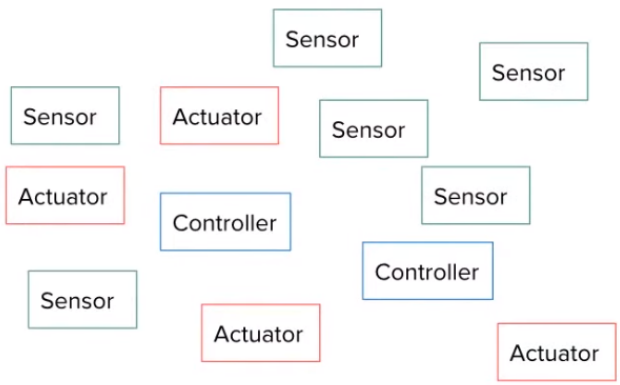
13

When?



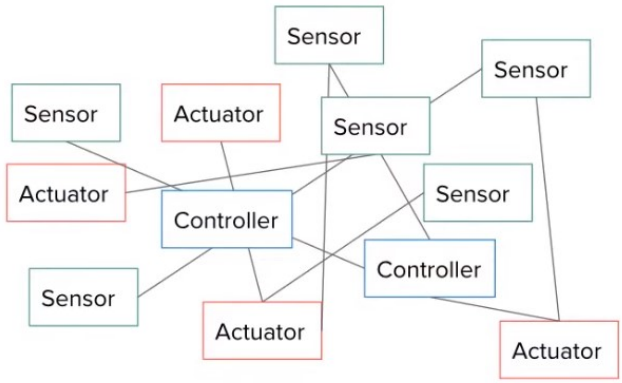
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When?



15

When?



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When?

- For which of the following should you use ROS?
- A robot that will just open one door with one motor.
- A mobile Robot which you want to control with a GPS and a camera

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When?

- So, for a robot which needs lots of communication between it's components

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What does it provide?

- Code separation and Communication tools
- Plug and Play libraries

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Example

- Imagine making a Robotic Arm!



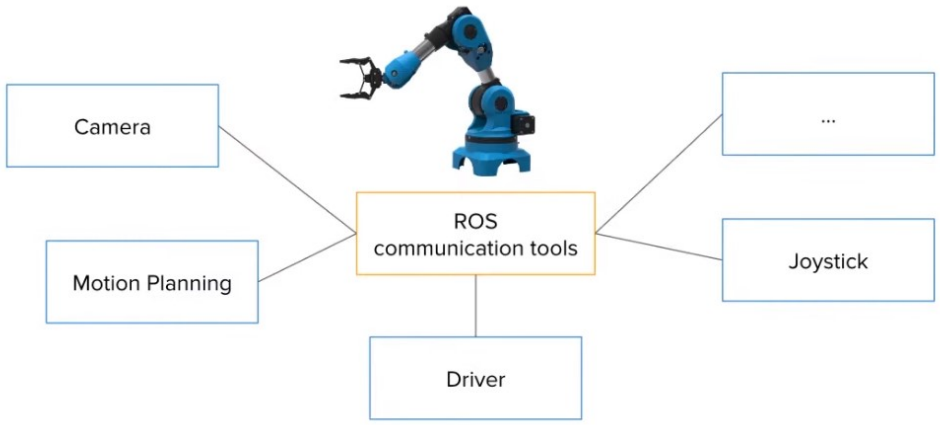
20

Example



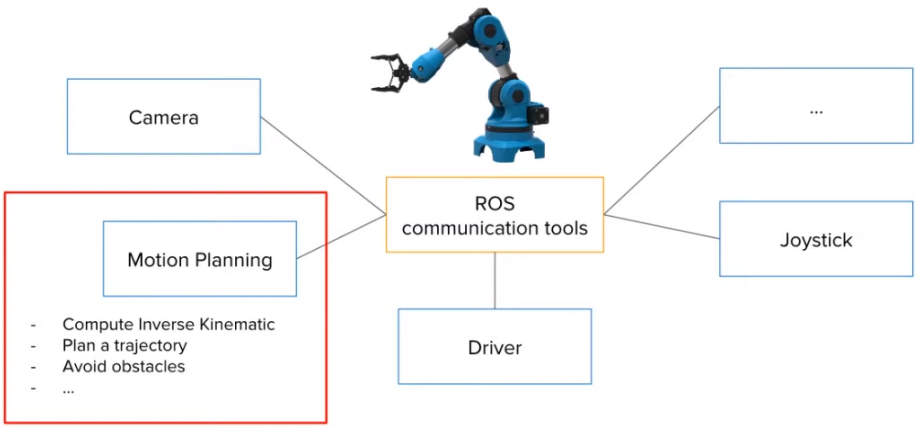
21

Example



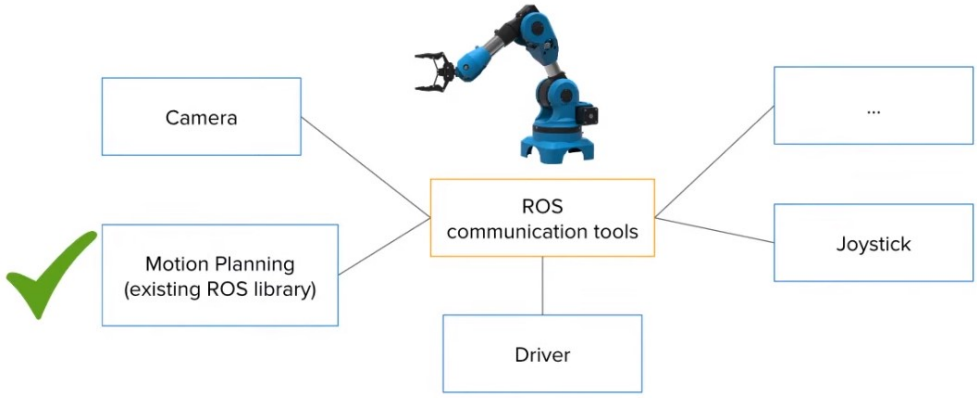
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Example



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Example



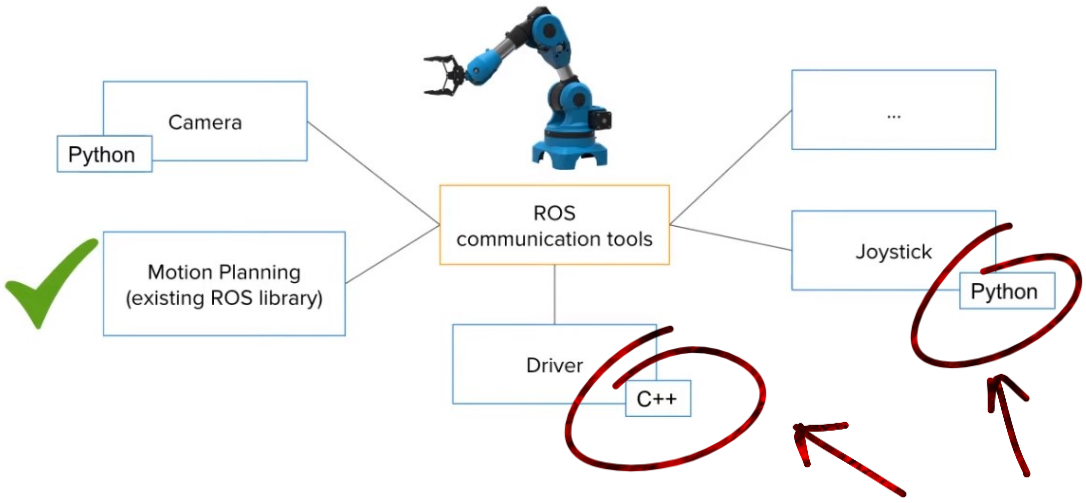
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Features

- Modular
- Scalable
- Language Agnostic!

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Example



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Features

- Open Source
- Active Community

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Major Versions

- Two!
- ROS1
- ROS2

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Official Documentation

- https://docs.ros.org/



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ROS 1

- Linux Only: Ubuntu, Debian
- C++, Python

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Installation Steps – 1

- Download Linux: Ubuntu 16.04
- https://ubuntu.com/download/desktop

- Install as a dual boot
- OR
- Install VmWare Workstation or Virtual Box or any Virtual Machine
- Then Install Linux on it

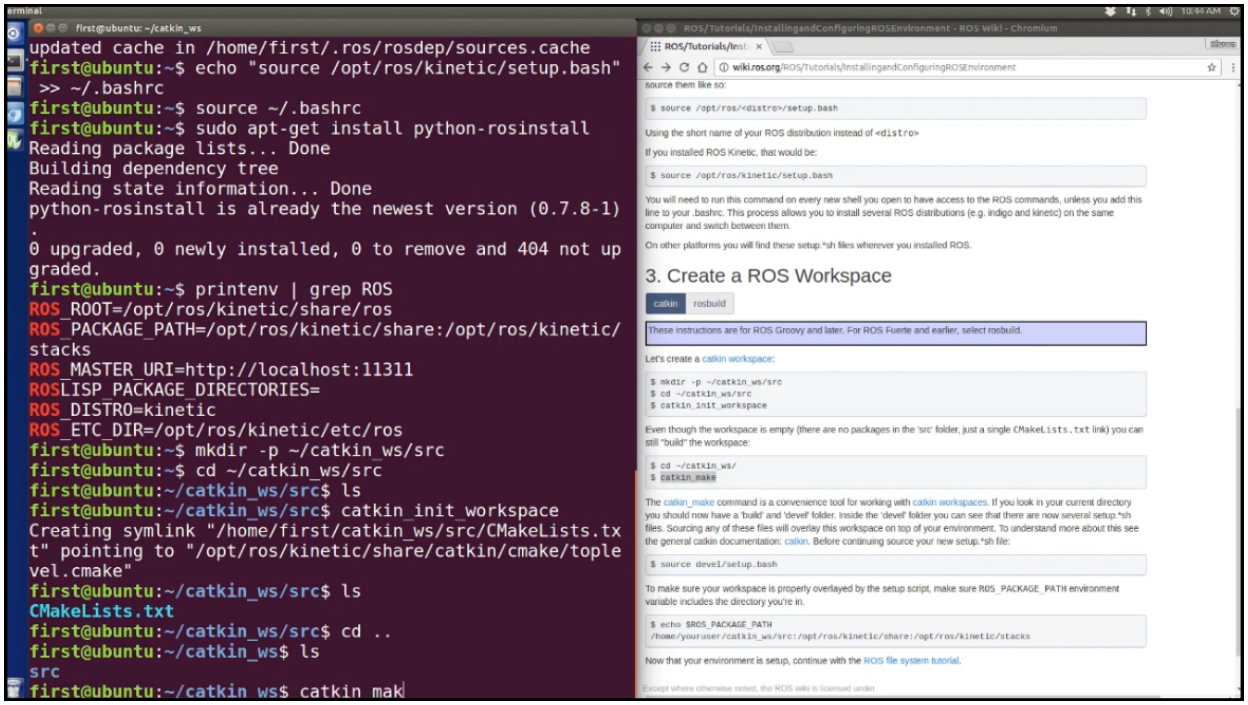
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Installation Steps – 2

- Open your terminal in the linux.
- Go to official ROS installation page:
- https://wiki.ros.org/ROS/Installation

- Glimpse of an Example in next slide

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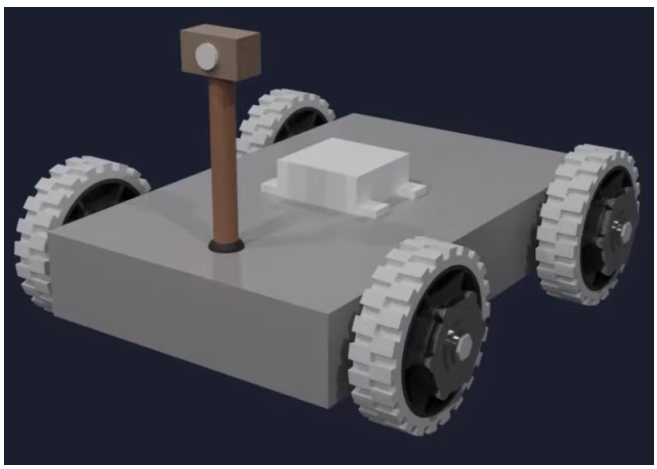
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ROS concepts (ROS 2)

- Let's take the example of a Mobile robot in the further concepts
- It has 4 wheels

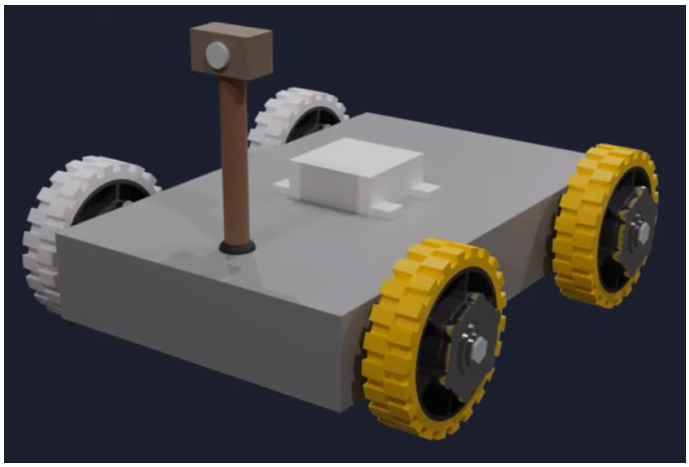
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Example



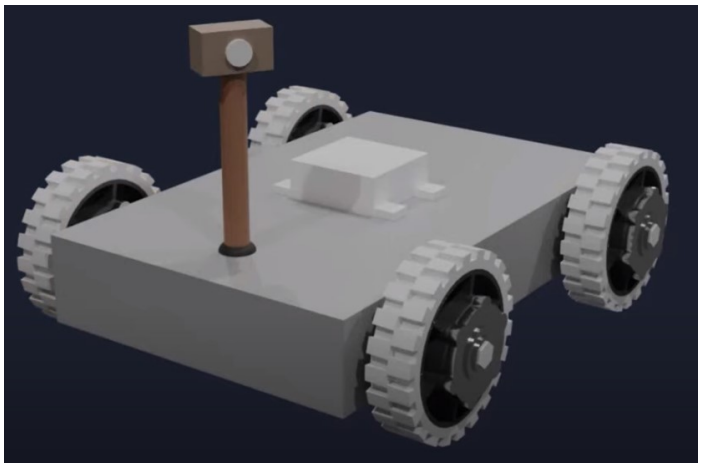
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Two motors



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Motorized Camera



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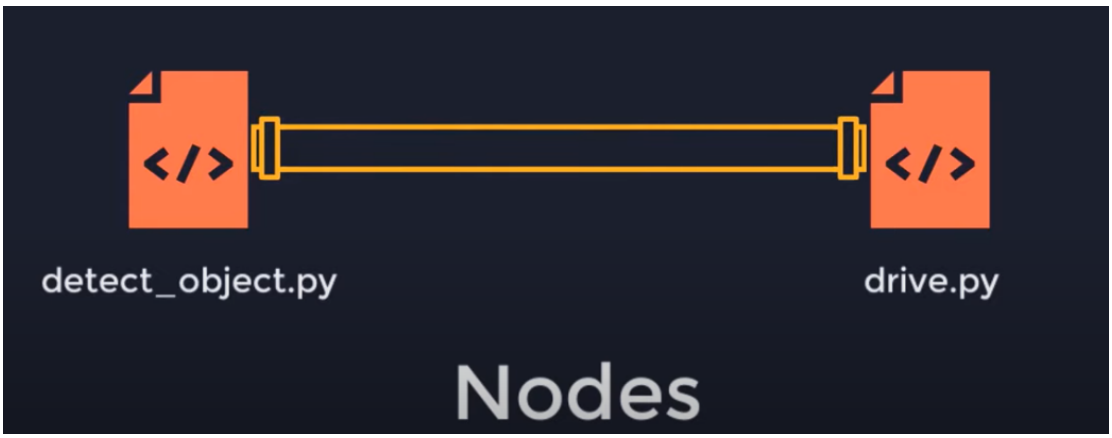
Data Distribution Service

- Communication pipeline
- 3 ways through which nodes communicate

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Nodes

Executed Code



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Data Distribution Service

- Communication pipeline
- 3 primary ways through which nodes communicate
  - Publishers/Subscribers
  - Service
  - Actions

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Publishers/Subscribers



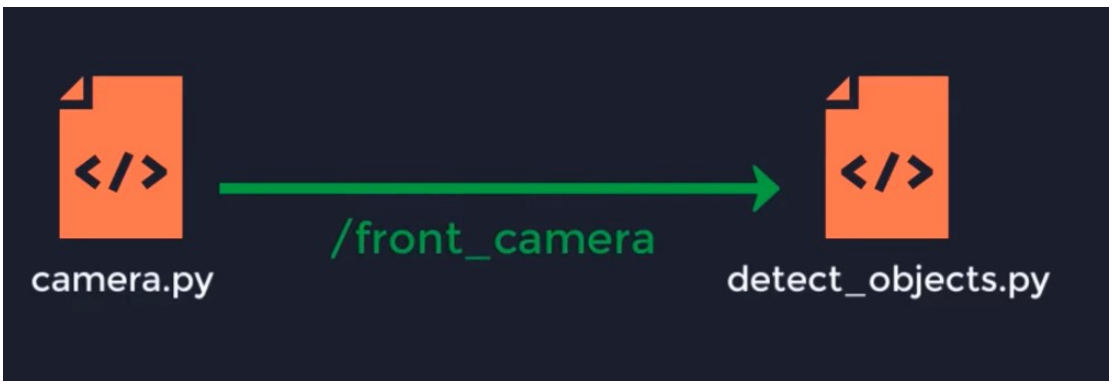
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They send information on a topic



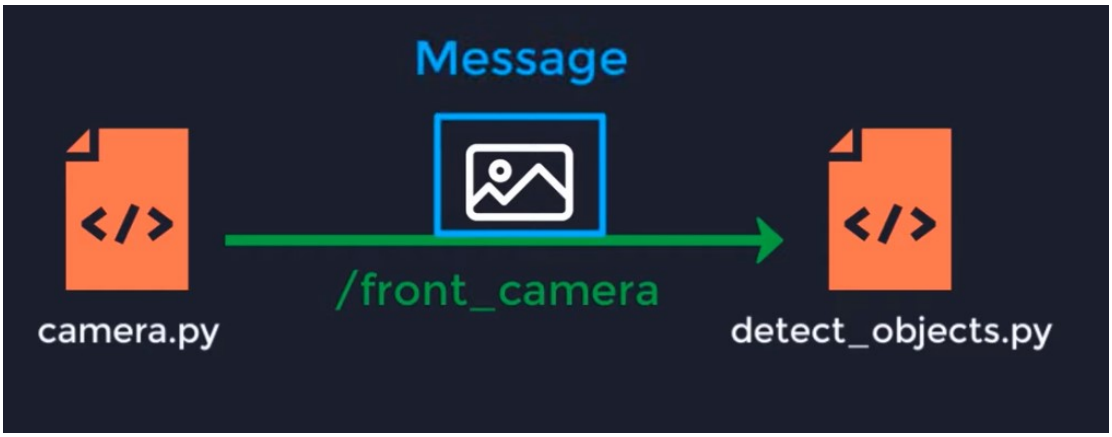
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E.G Topic



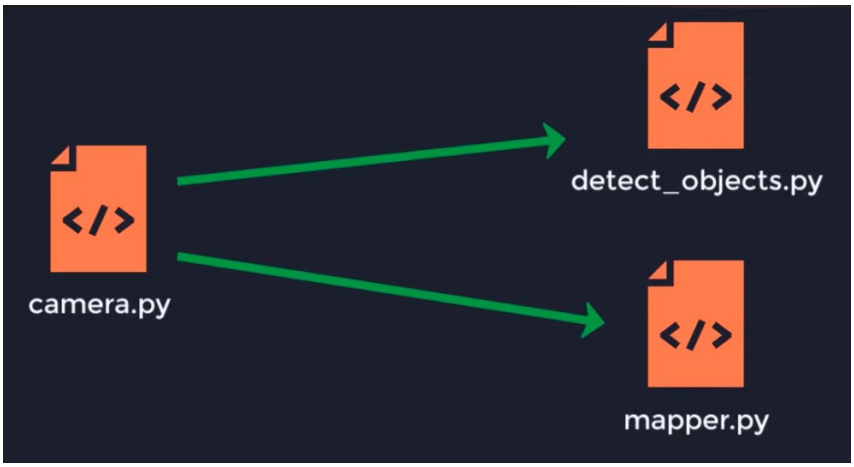
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Information sent is called Message



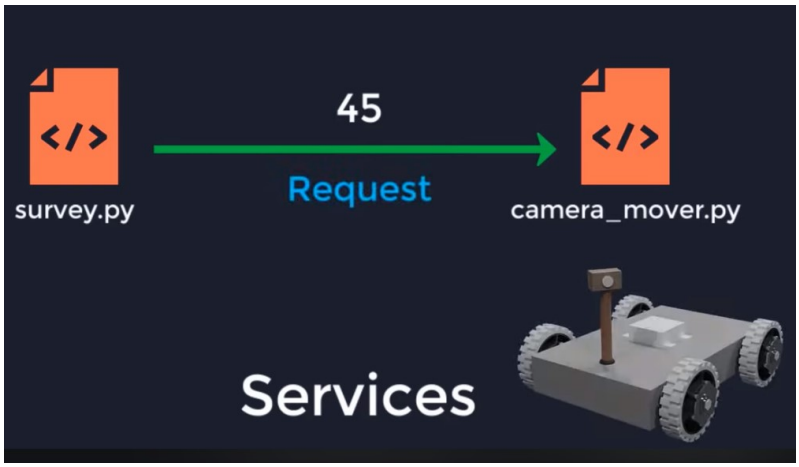
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Can be multiple Subscribers



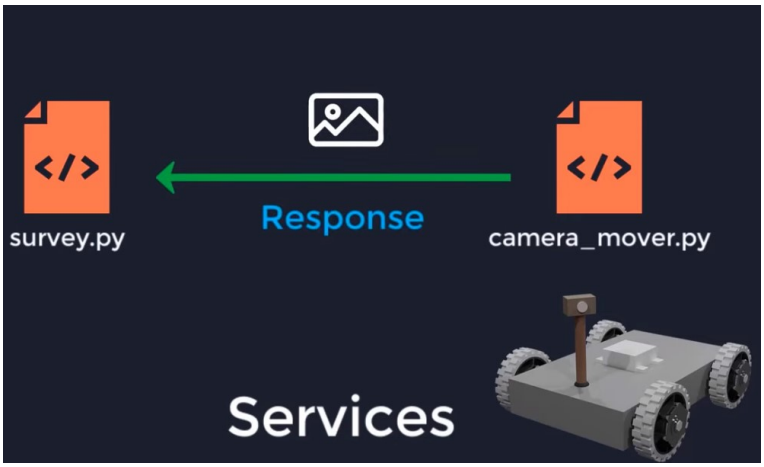
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Service



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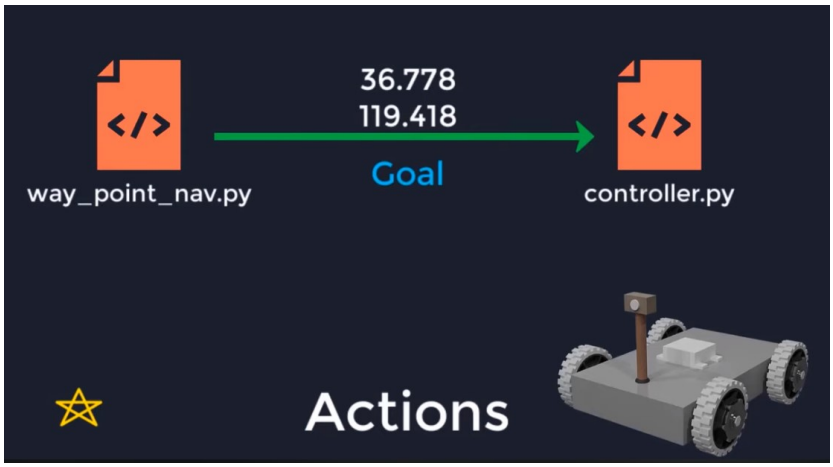
Service



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Actions

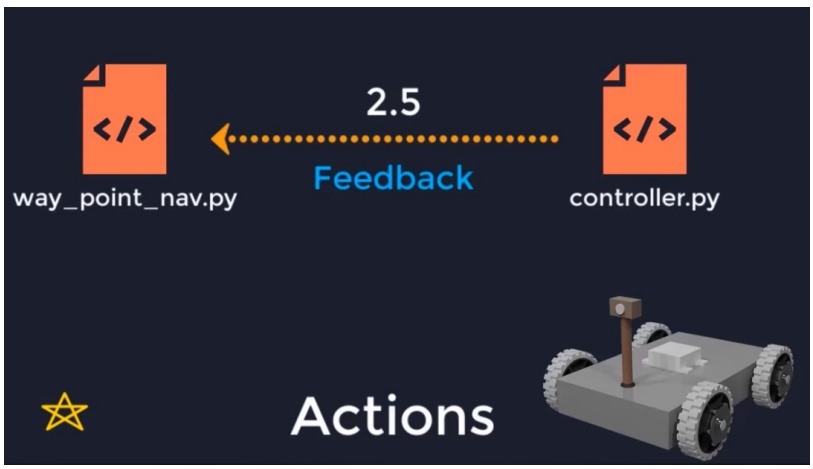
E.G Our node wants to command our robot to travel to certain longitude and latitude and take a picture.



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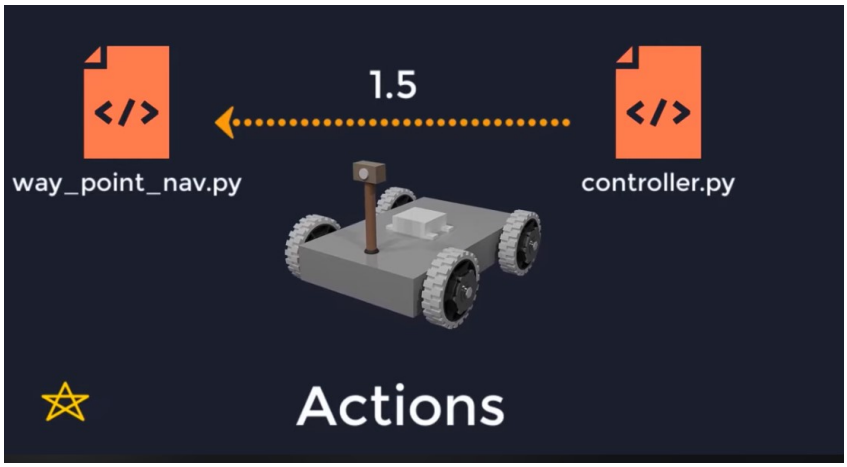
Actions

It will send progress updates to client until goal is reached e.g how far away is the goal.



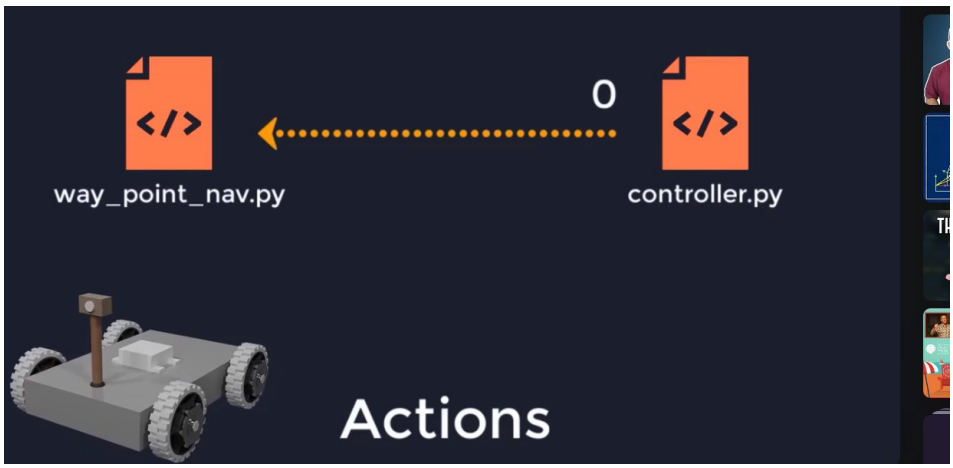
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Actions



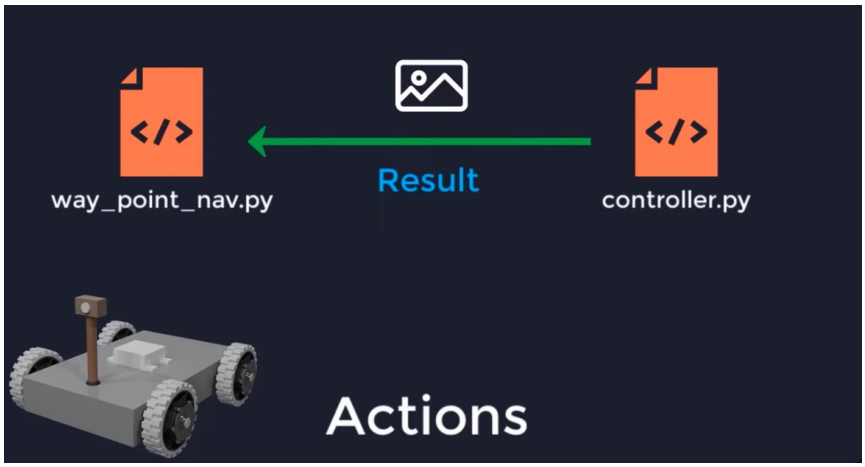
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Actions



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Actions



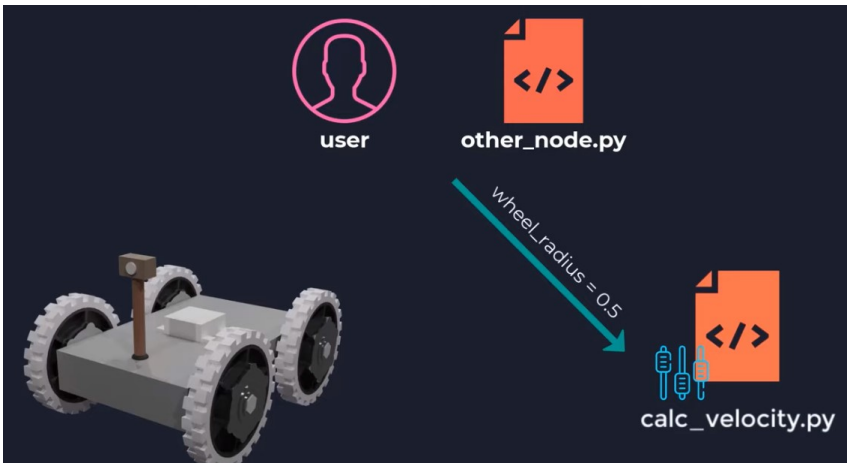
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Node Parameters

- Node Parameters
- E.g wheel size is changed.
- Just change the parameter instead of changing the whole code.

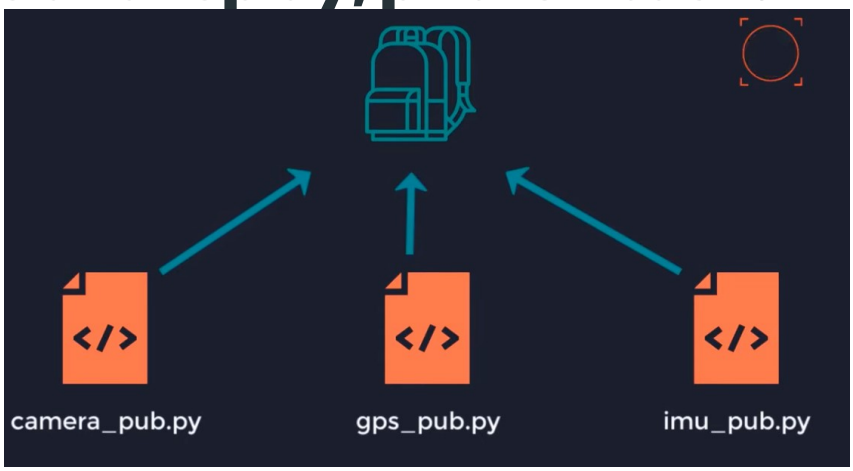
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Node Parameters



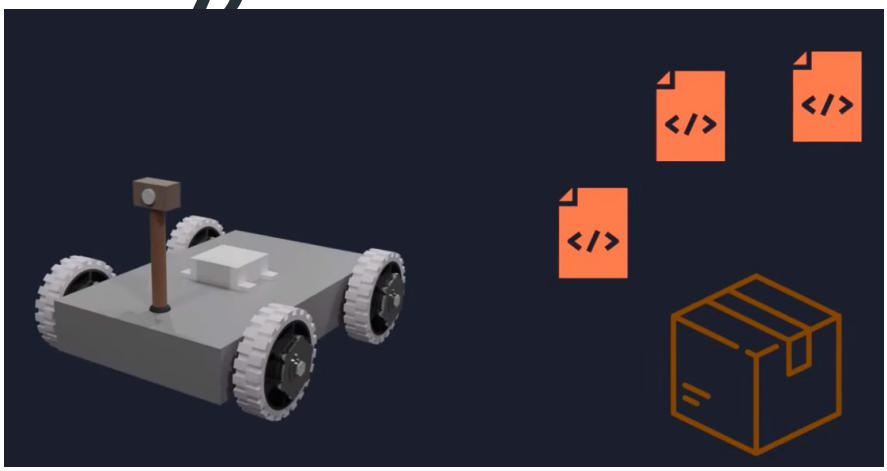
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Bag files: Subscribe to multiple topics Save and replay, publish back.



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Packages (pack all in one for reusability)



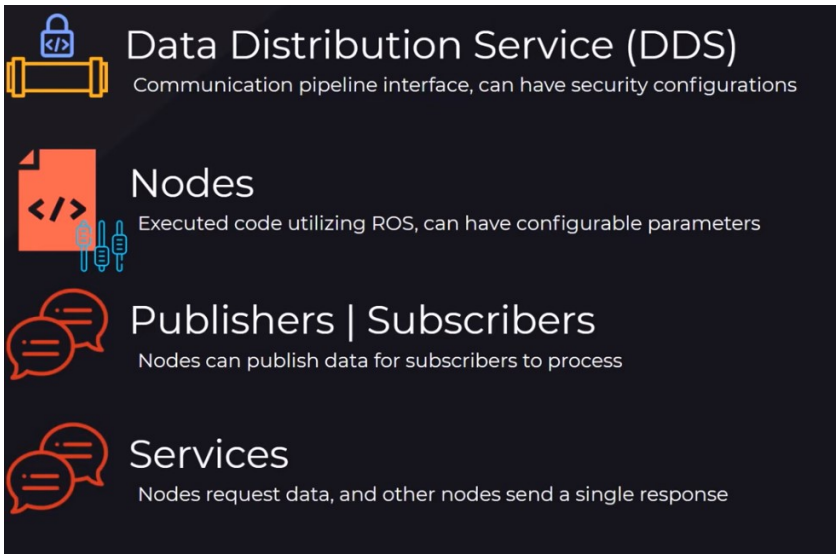
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More features

- Bag files
- Packages
- Cross platform support

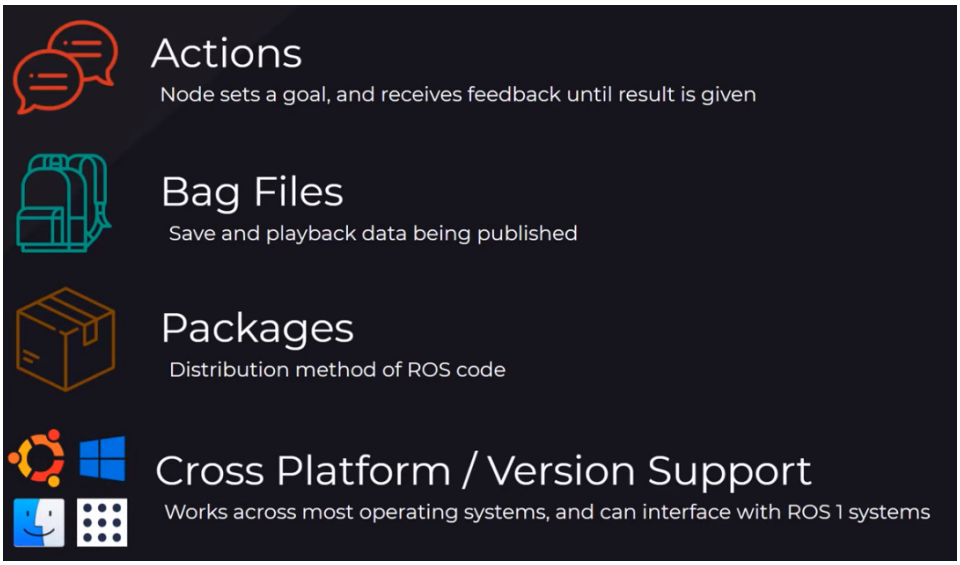
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Summary

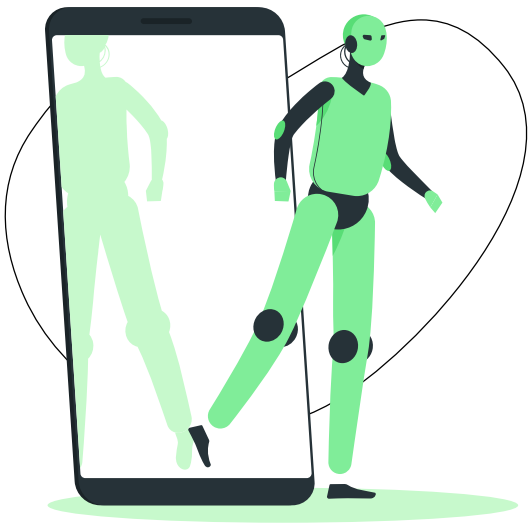


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Summary



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Thanks!

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